COMPSCI 572.01 (NLP)

Post-training 1:
(Instruction Tuning, RLHF)

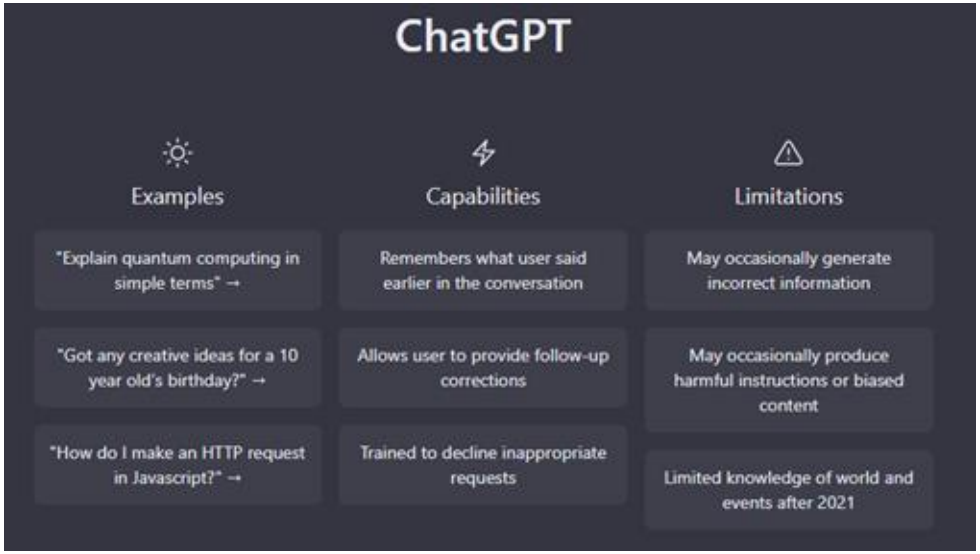
Slides based on those of Tatsunori Hashimoto, Greg Durrett

Overview

- Introduction
- Instruction tuning
  - Approach
  - Evaluation
  - Limitations
- RLHF
  - Reward modeling
  - Policy gradient methods

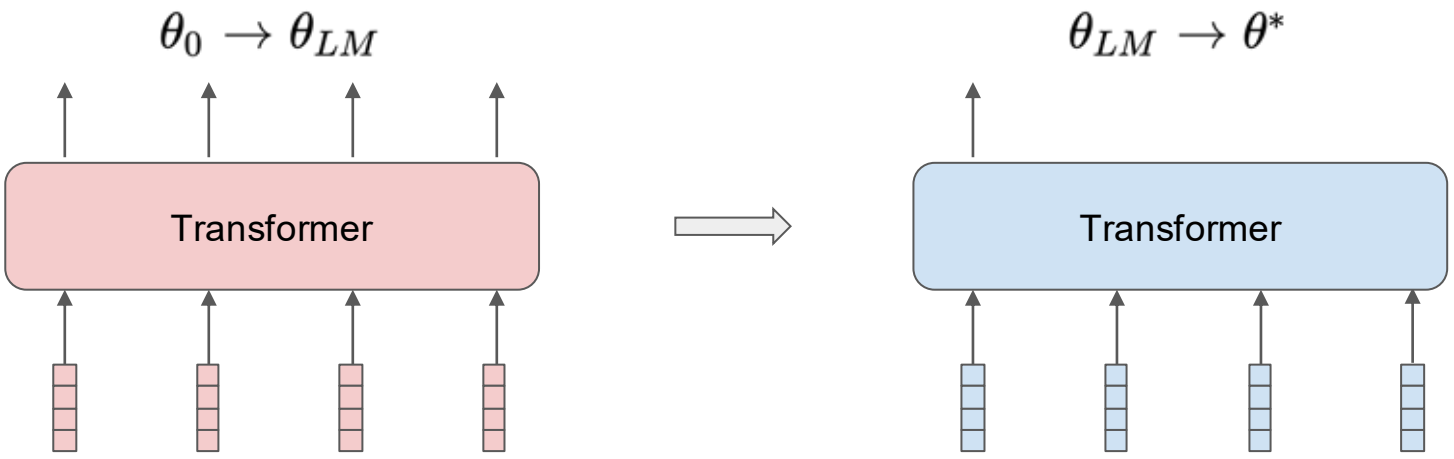
Language models to general-purpose assistants

- We want models which predict P(answer | ~~prompt~~ instruction)
- Language modeling alone does not teach the model to do this



Recall: Pretrain then finetune

- **Idea:** Let's finetune the models to do what we care about

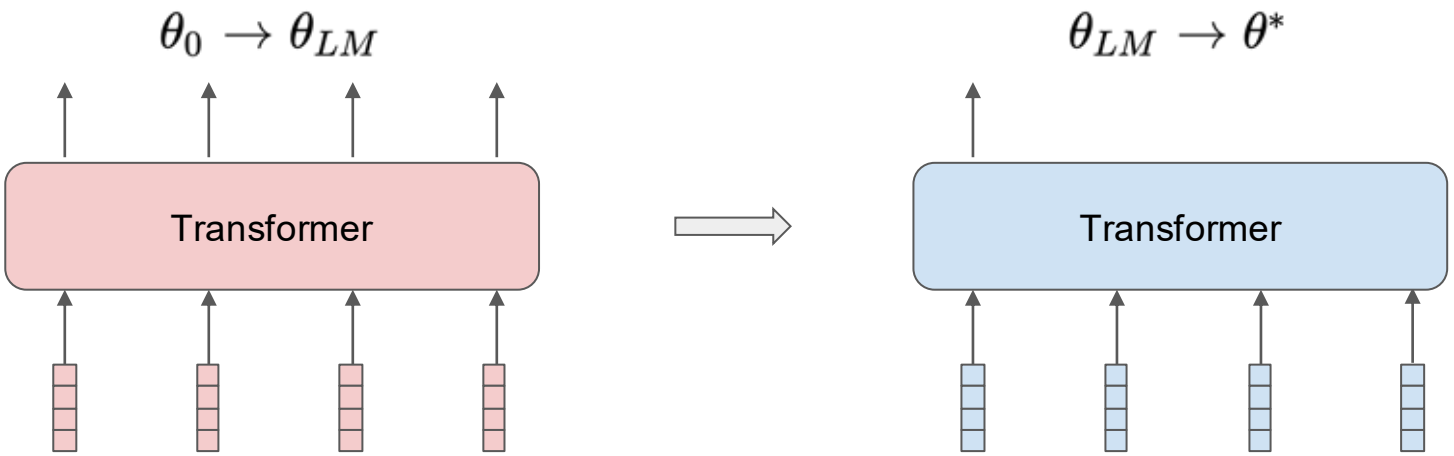


1. **Pretrain** on language modeling (lots of unlabeled data)

2. **Finetune** on a **single** task (limited labeled data)

Today: Scaling up finetuning

- **Idea:** Let's finetune the models to do what we care about



1. **Pretrain** on language modeling (lots of unlabeled data)

2. **Finetune** on a **many** tasks (**lots of** labeled data)

Types of labeled data

- **Supervised data:** used for instruction tuning
  - *Input* ***x*** = “who was the ruler of France in 1830?”
    Gold output ***y**** = “Charles X, Louis-Phillippe I”

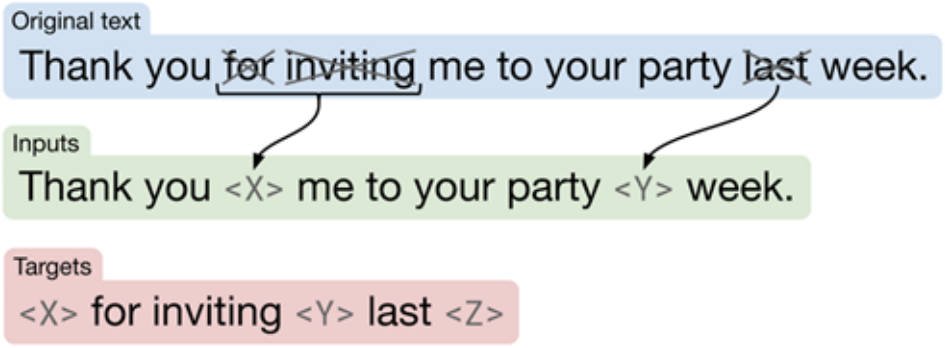
- **Preferences:** used for reinforcement learning (RLHF)
  - *Input* ***x*** = “who was the ruler of France in 1830?”
    Preferred ***y+*** = “Charles X, Louis-Phillippe I”
    Unpreferred ***y-*** = “Louis-Phillippe I”

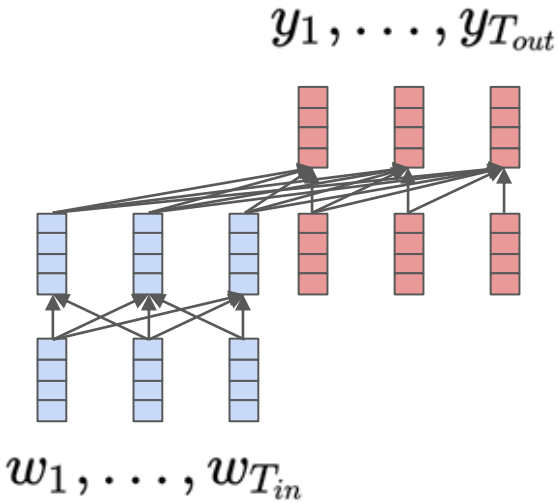
  - *Preferred* ***y+*** = “Charles X was the ruler until August 2, after that Louis-Phillippe I”
    Unpreferred ***y-*** = “Charles X, Louis-Phillippe I”

Recap: Encoder-Decoder Pretraining

Or we can again use *masked* language modeling (a.k.a *Denoising*)

- Replace spans in the input with special tokens
- Predict those spans at the output



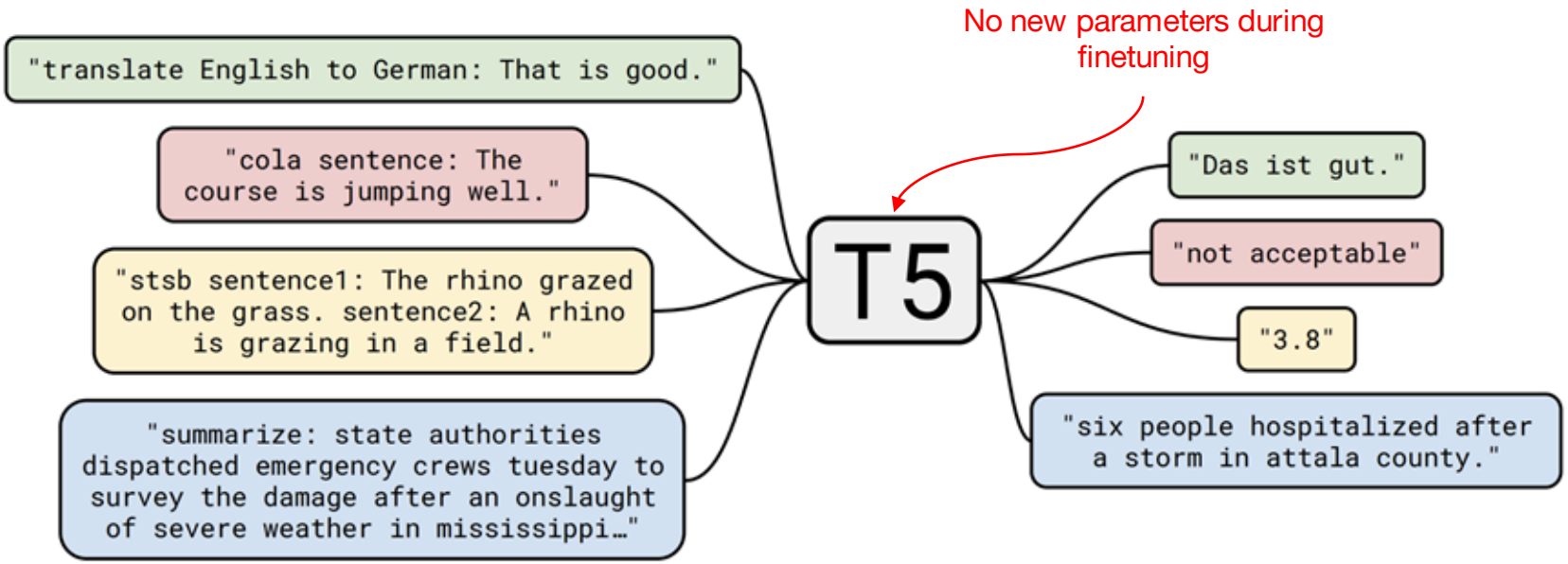


Exploring the limits of text-to-text transfer learning Raffel et al; 2019

Recap: Encoder-decoder Finetuning

Text-To-Text Transfer Transformer (T5)

Formulate all tasks as a sequence-to-sequence problem



Exploring the limits of text-to-text transfer learning Raffel et al; 2019

Instruction Tuning

T0

Published as a conference paper at ICLR 2022

MULTITASK PROMPTED TRAINING ENABLES ZERO-SHOT TASK GENERALIZATION

Victor Sanh* Hugging Face
Albert Webson* Brown University
Colin Raffel* Hugging Face
Stephen H. Bach* Brown & Snorkel AI

Lintang Sutawika BigScience
Zaid Alyafeai KFUPM
Antoine Chaffin IRISA & IMATAG
Arnaud Stiegler Hyperscience
Teven Le Scao Hugging Face

Arun Raja I²R, Singapore
Manan Dey SAP
M Saiful Bari NTU, Singapore
Canwen Xu UCSD & Hugging Face
Urmish Thakker SambaNova Systems

Shanya Sharma Walmart Labs
Eliza Szczechla BigScience
Taewoon Kim VU Amsterdam
Gunjan Chhablani BigScience
Nihal V. Nayak Brown University

Debajyoti Datta University of Virginia
Jonathan Chang ASUS
Mike Tian-Jian Jiang ZEALS, Japan
Han Wang NYU
Matteo Manica IBM Research

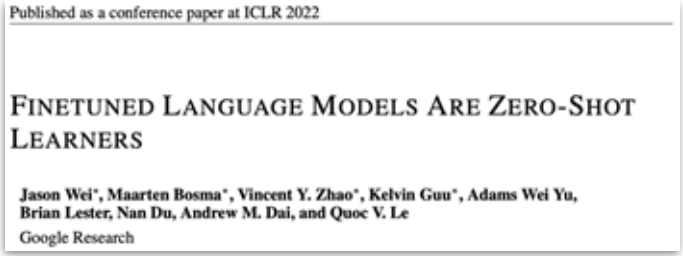
Sheng Shen UC Berkeley
Zheng-Xin Yong Brown University
Harshit Pandey BigScience
Michael McKenna Parity
Rachel Bawden Inria, France

Thomas Wang Inria, France
Trishala Neeraj BigScience
Jos Rozen Naver Labs Europe
Abheesht Sharma BITS Pilani, India
Andrea Santilli University of Rome

Thibault Fevry BigScience
Jason Alan Fries Stanford & Snorkel AI
Ryan Teehan Charles River Analytics
Tali Bers Brown University

Stella Biderman Booz Allen & EleutherAI
Leo Gao EleutherAI
Thomas Wolf Hugging Face
Alexander M. Rush Hugging Face

Flan-T5

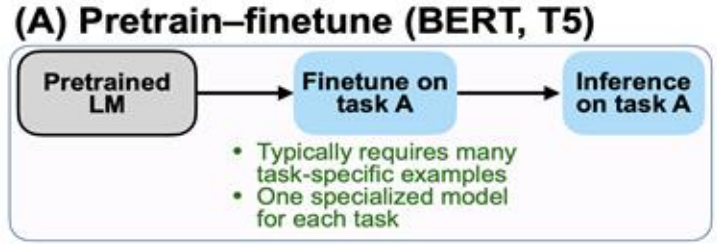


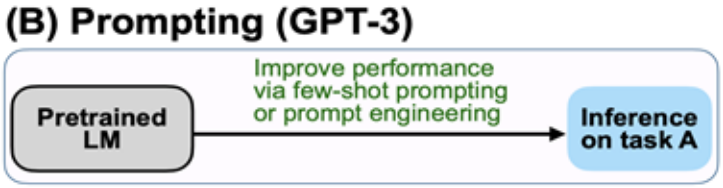
Published as a conference paper at ICLR 2022

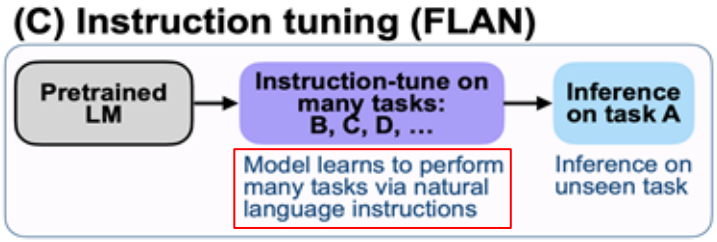
FINETUNED LANGUAGE MODELS ARE ZERO-SHOT LEARNERS

Jason Wei*, Maarten Bosma*, Vincent Y. Zhao*, Kelvin Guu*, Adams Wei Yu, Brian Lester, Nan Du, Andrew M. Dai, and Quoc V. Le

Google Research

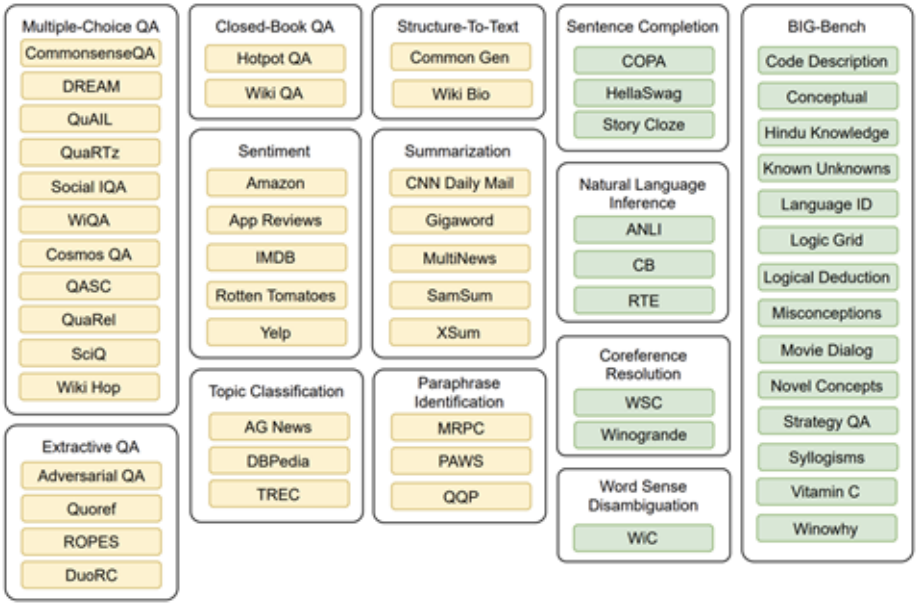


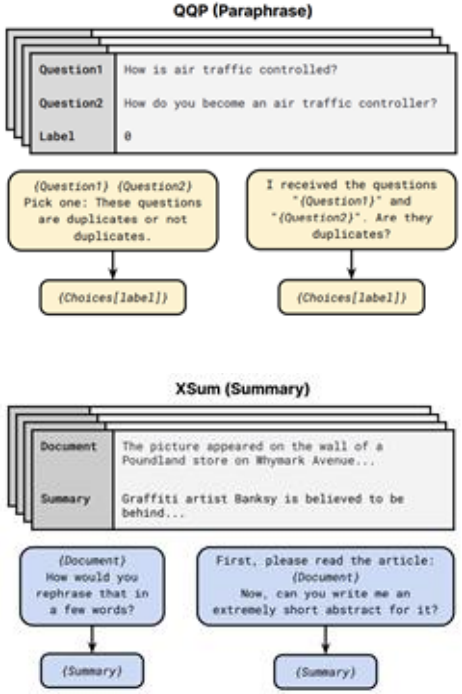




Instruction Tuning

Both these papers collected data by crowd-sourcing templates for existing NLP datasets

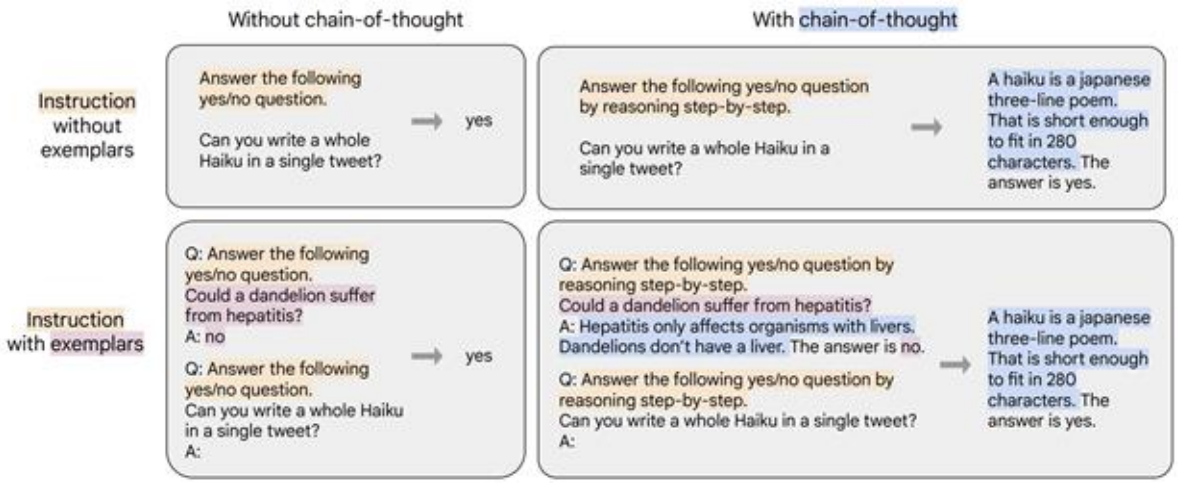




https://arxiv.org/abs/2110.08207

Flan-PaLM

- As always, scaling up data and model size helps!
- **Flan-PaLM**: 1.8K tasks, 540B parameter model
  - Instruction tuned with and without examples and with and without chain-of-thought



https://arxiv.org/abs/2210.11416

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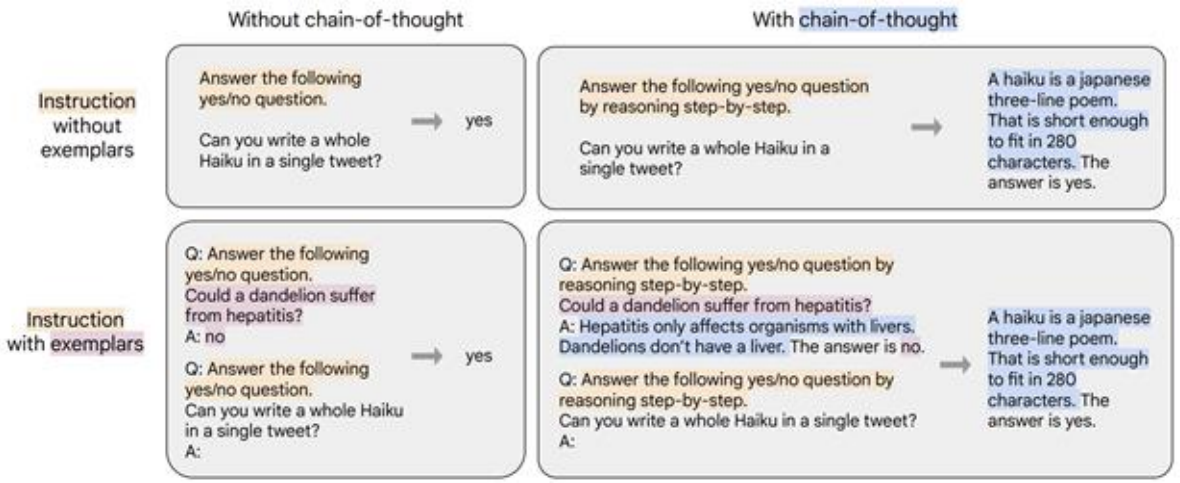
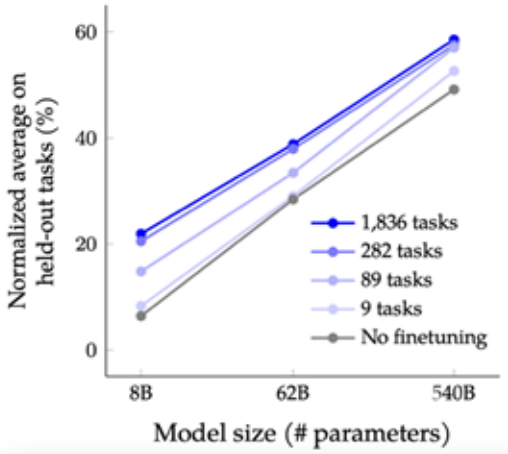




https://huggingface.co/docs/transformers/en/model_doc/flan-t5

Evaluating instruction-following

- Several attempts at creating benchmarks which test for diverse capabilities

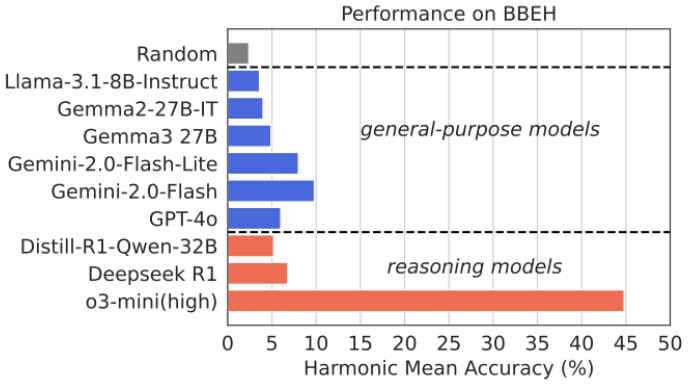
Q: What movie does this emoji describe? 👧🐟🐠🐡

1b: the emoji movie
2b: the emoji movie
4b: the emoji for a baby with a fish in its mouth
8b: the emoji movie
27b: the emoji is a fish
128b: finding nemo

BIG-Bench (Srivastava et al, 2023)
>200 language tasks with eval metrics

Evaluating instruction-following

"recent advances in LLMs have led to saturation on BIG-Bench, and its harder version BIG-Bench Hard (BBH)…"



BoardgameQA

A few players are playing a boardgame. The current state of the game is as follows. The bee has a football with a radius of 15 inches. The chihuahua has a smoothie. [...]
Rule1: Anyone who [...] has to pay $$$ to the beaver.
Rule2: If the stork [...], then it does not pay $$$ to beaver.
Rule 2 is preferred over Rule 1. If a rule is preferred over the other, it means whenever both of them can be applied to derive new conclusions and those conclusions contradict with each other, we should go with the conclusion from the rule with higher preference.
What is the truth value of the statement: "Does the finch shout at the mermaid?"

BIG-Bench Extra Hard (Kazemi et al, 2025)
From Google DeepMind

Evaluating instruction-following

- Several attempts at creating benchmarks which test for diverse capabilities

Find all c in $\mathbb{Z}_3$ such that $\mathbb{Z}_3[x]/(x^2 + c)$ is a field.
(A) 0 **(B) 1** (C) 2 (D) 3

Figure 14: An Abstract Algebra example.

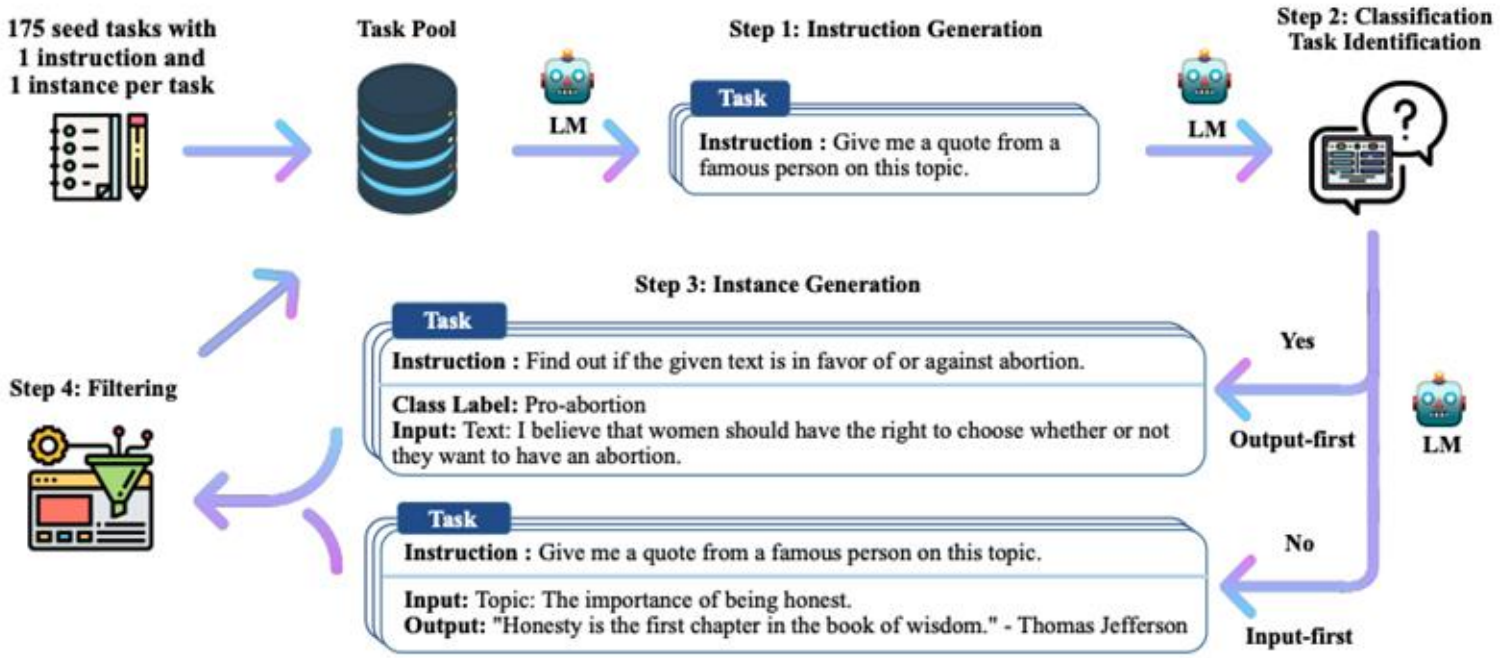
What is the embryological origin of the hyoid bone?
(A) The first pharyngeal arch
(B) The first and second pharyngeal arches
(C) The second pharyngeal arch
(D) The second and third pharyngeal arches

Figure 15: An Anatomy example.

MMLU (Hendrycks et al, 2021)
57 multiple choice tasks (topics)

Is human-labeled data necessary for instruction tuning?

No.

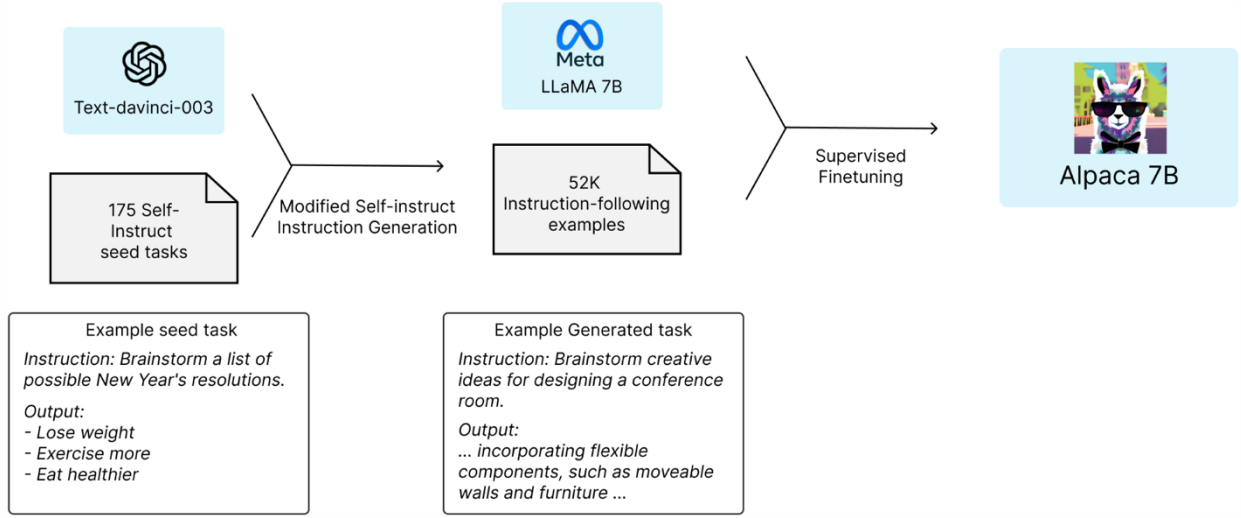


Self-instruct, Wang et al (2023)

Alpaca

Alpaca: A Strong, Replicable Instruction-Following Model

Stanford Alpaca



Alpaca, Taori, Gulrajani, Zhang, Dubois, Li et al

Limitations

Instruction tuning is straightforward and improves generalization to unseen tasks, but has some limitations:

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Instruction tuning is straightforward and improves generalization to unseen tasks, but has some limitations:

1. Some tasks do not have a clearly defined correct answer
   "Got any creative ideas for a 10-year old's birthday?"

2. Supervised fine-tuning penalizes all token mistakes equally
   Gold: *"Charles X was the ruler of France in 1830"*
   Option 1: *"~~Charles~~ Philipp X was the ruler of France in 1830"*
   Option 2: "Charles X was the ~~ruler~~ king of France in 1830"

In other words, instruction tuning doesn't align with fine-grained human preferences

Optimizing for human preferences

- Let's assume, given an instruction, we have a way to measure a **reward** for the quality of an output from the LM p_θ

TITLE: Screwed up with boss... what should I do?
I'm 20 f, my boss is around 50 years old, also f.
So I have two jobs, and the schedules for both jobs are made on a weekly basis. One of my jobs I have had for three years, the other one I have had for a month and a bit. I forgot to give my schedule from one job to my boss at my other job, and so I was not scheduled for this week. I didn't realize why I had not been put on the schedule until now.

s1 ~ p_θ

s2 ~ p_θ

Summary: I forgot to give my boss my weekly schedule for one of my jobs, and so I was not scheduled this week. I royally screwed up. What can I do to redeem myself?

R(s1) = 5

Summary: I royally screwed up and I don't know what to do.

R(s2) = 3

- Our goal is to maximize the expected reward $\mathbb{E}_{s\sim p_\theta}[R(s)]$

Recall: The policy-perspective on seq2seq

In RL and IL, we are interested in learning a "policy."

- A policy consumes the "state" of the world, and gives a distribution over next-actions.

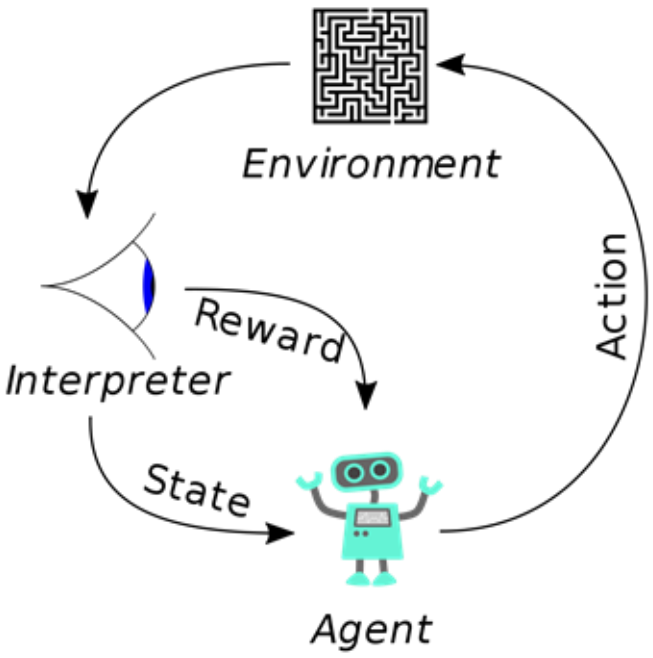

Figure credit: Megajuice, Wikipedia

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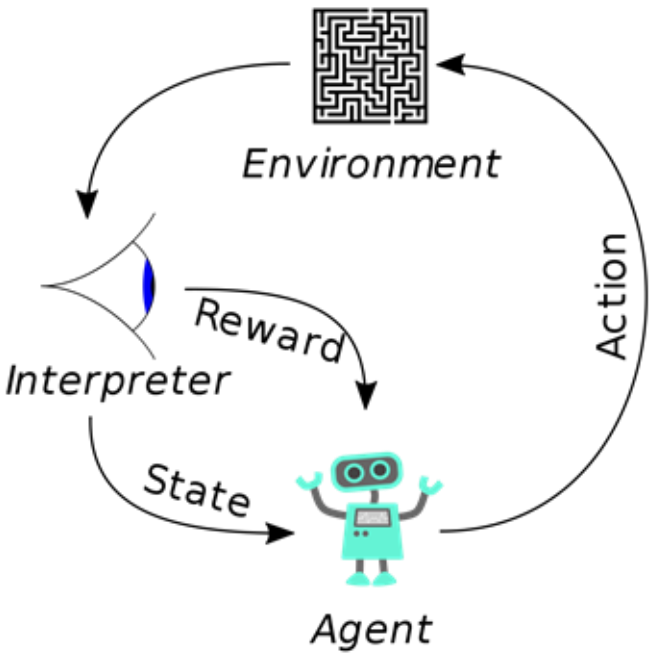
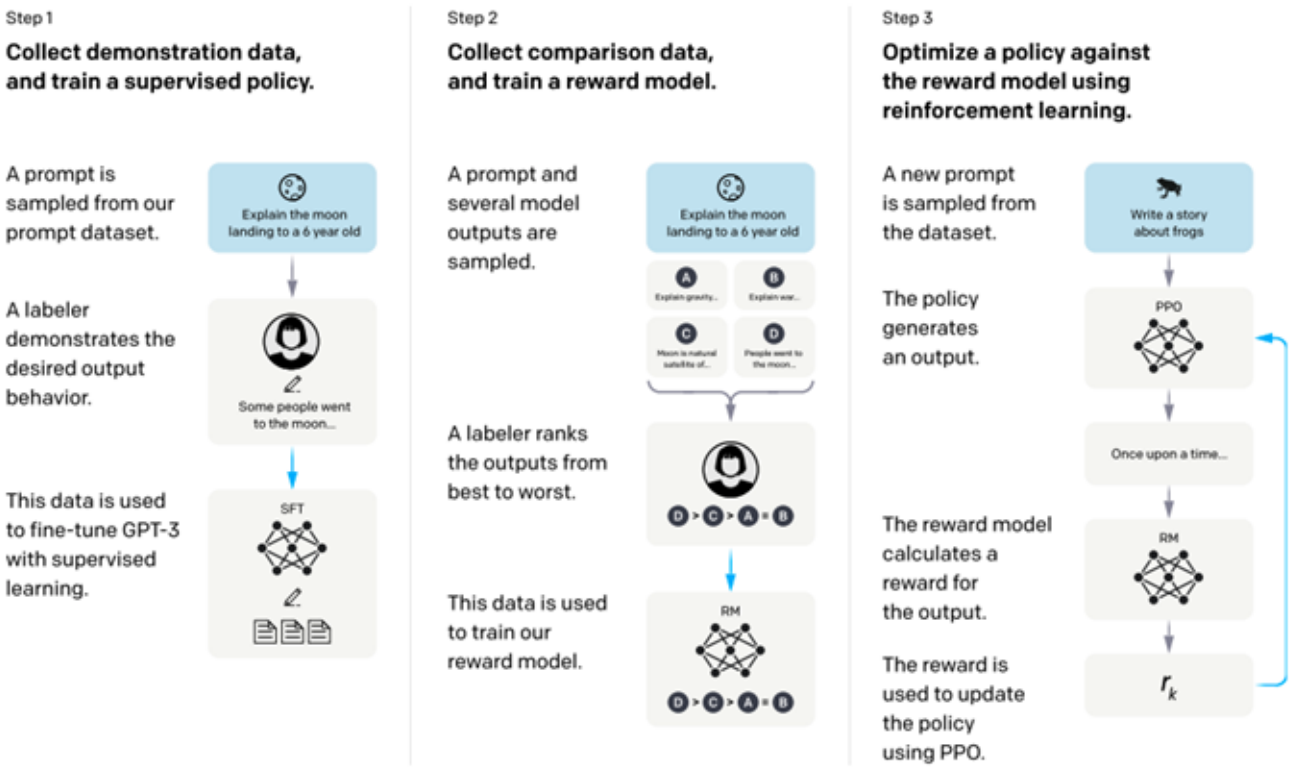


These ideas were not very successful for training towards an automatic eval metric.

But, newer algorithms and large-scale human feedback has led to a resurgence in interest in RL training of LMs.

Figure credit: Megajuice, Wikipedia

Reinforcement learning from human feedback (RLHF) Pipeline

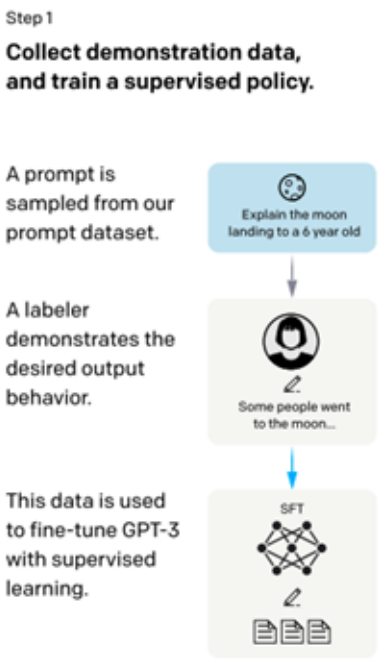


InstructGPT, Ouyang et al, 2022

RLHF Pipeline Step 1: Supervised Finetuning

- This is the same as instruction tuning
- But instead of repurposing existing NLP datasets, authors used **diverse instruction prompts from the GPT-3 API**

Use-case	Prompt
Brainstorming	List five ideas for how to regain enthusiasm for my career
Generation	Write a short story where a bear goes to the beach, makes friends with a seal, and then returns home.
Rewrite	This is the summary of a Broadway play: """ {summary} """ This is the outline of the commercial for that play: """



InstructGPT, Ouyang et al, 2022

RLHF Pipeline Step 2: Reward Modeling

- Human judgments are noisy and miscalibrated across examples
- Instead of direct ratings, more common to use **pairwise preferences**
- Data can be collected by ranking K outputs
- Then, for each pair of outputs, train the reward model to maximize (Bradley-Terry model):

$$P(s^+ \succ s^-) = \frac{\exp(r_\phi(s^+))}{\exp(r_\phi(s^+)) + \exp(r_\phi(s^-))}$$

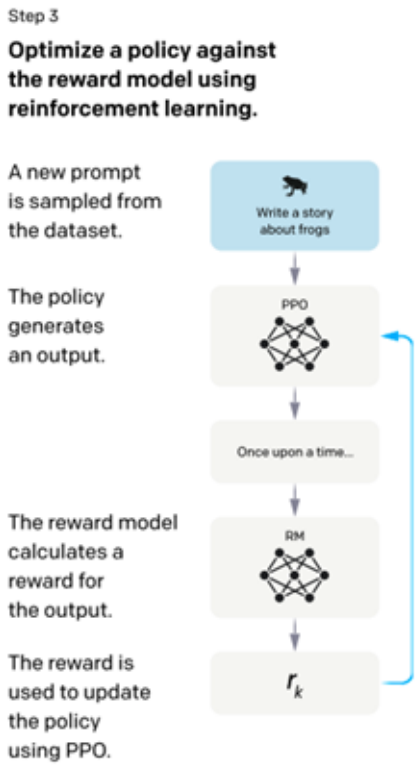
"Winning" output — "Losing" output — Reward model

InstructGPT, Ouyang et al, 2022

RLHF Pipeline Step 3: Reinforcement learning

- We now have a way to measure the reward $r_\phi()$
- How do we train the LM parameters θ to maximize $\mathbb{E}_{s\sim p_\theta}[r_\phi(s)]$?
- Typically, SGD:

$$\theta_{t+1} \leftarrow \theta_t + \eta \nabla_\theta \mathbb{E}_{s\sim p_\theta}[r_\phi(s)]$$



InstructGPT, Ouyang et al, 2022

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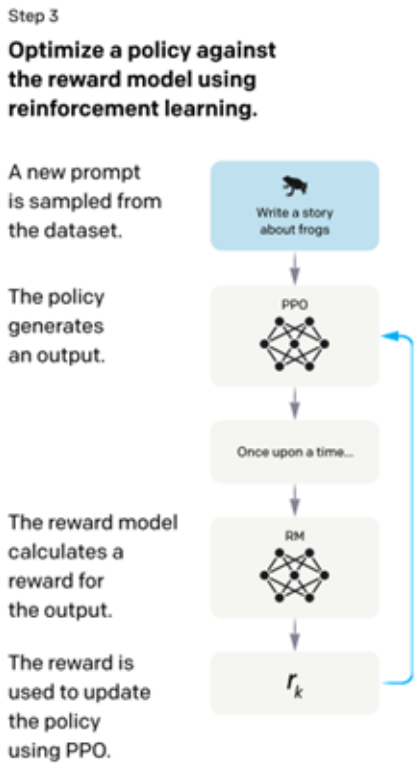
How do we estimate this expectation?

How do we compute the gradient wrt to the non-differentiable reward?



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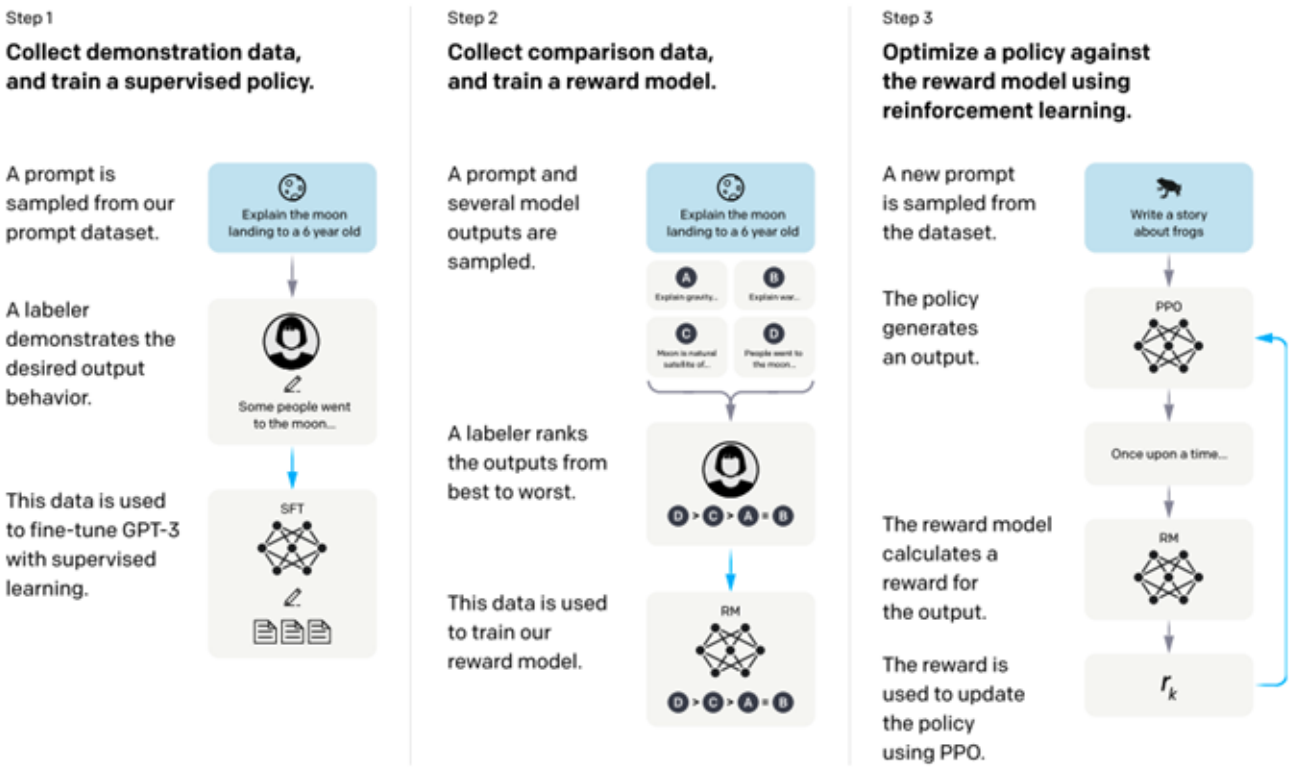

InstructGPT, Ouyang et al, 2022

RLHF Pipeline Step 3: Reinforcement learning

- We want to maximize:

$$\mathbb{E}_{s\sim p_\theta}[r_\phi(s)]$$

wrt the LM parameters θ

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How do we estimate this expectation?

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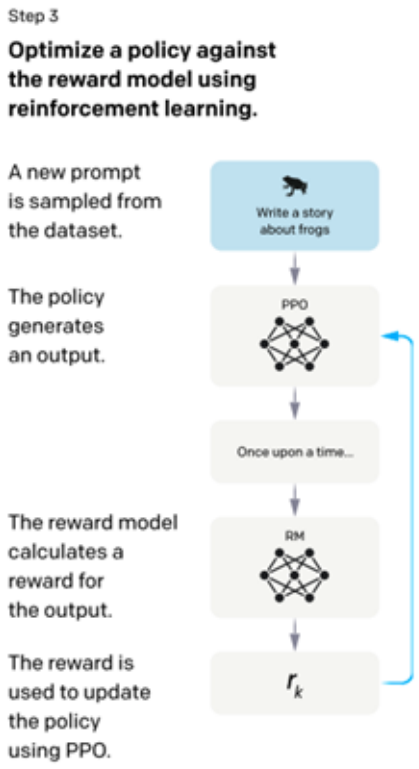
⇒ Policy gradient methods!



InstructGPT, Ouyang et al, 2022

Policy gradient (REINFORCE)

- We want to compute:

$$\nabla_\theta \mathbb{E}_{s\sim p_\theta}[r_\phi(s)] = \nabla_\theta \sum_s p_\theta(s) r_\phi(s) = \sum_s \nabla_\theta p_\theta(s) r_\phi(s)$$

Definition of expectation

Linearity of gradient

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Definition of expectation

Linearity of gradient

- Log-derivative trick:

$$\nabla_\theta \log p_\theta(s) = \frac{1}{p_\theta(s)} \nabla_\theta p_\theta(s) \quad \Rightarrow \quad \nabla_\theta p_\theta(s) = p_\theta(s) \nabla_\theta \log p_\theta(s)$$

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- Plugging back in:

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Definition of expectation Linearity of gradient

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- Plugging back in:

$$\sum_s \nabla_\theta p_\theta(s) r_\phi(s) = \sum_s p_\theta(s) r_\phi(s) \nabla_\theta \log p_\theta(s)$$

This is an expectation

We have moved the gradient inside the expectation!

$$\boxed{= \mathbb{E}_{s\sim p_\theta}[r_\phi(s) \nabla_\theta \log p_\theta(s)]}$$

RLHF Pipeline Step 3: Reinforcement learning

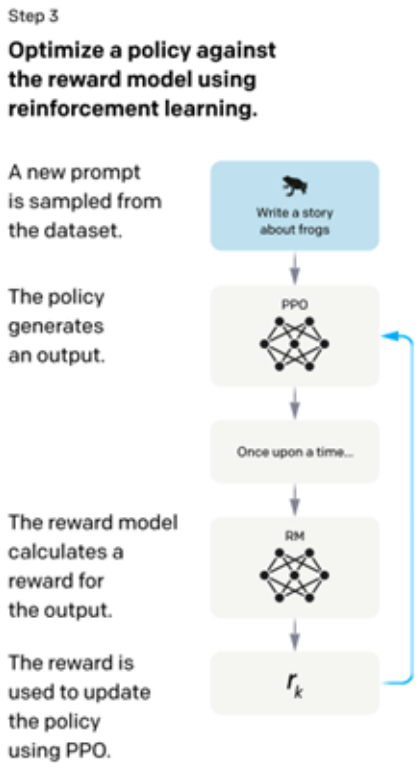
- We can estimate the expectation with samples

$$\nabla_\theta \mathbb{E}_{s\sim p_\theta}[r_\phi(s)] = \mathbb{E}_{s\sim p_\theta}[r_\phi(s)\nabla_\theta \log p_\theta(s)] \approx \frac{1}{m}\sum_{i=1}^{m} r_\phi(s_i)\nabla_\theta \log p_\theta(s_i)$$

- Update for the parameters θ:

$$\theta_{t+1} \leftarrow \theta_t + \eta \nabla_\theta \mathbb{E}_{s\sim p_\theta}[r_\phi(s)]$$

- If the reward is positive the update maximizes the (log)probability of the output (and vice versa)



InstructGPT, Ouyang et al, 2022

RLHF Pipeline Step 3: Reinforcement learning

- In practice, the reward which is maximized is:

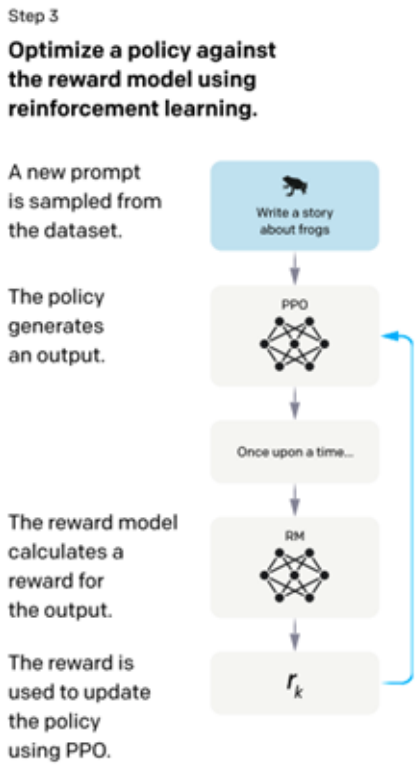
$$R(s) = r_\phi(s) - \lambda D_{KL}\left(p_\theta(s) \| p^{SFT}(s)\right)$$

KL-divergence

Fine-tuned LM (Step 1)

- Important to regularize the LM to stay close to the finetuned / pretrained LM

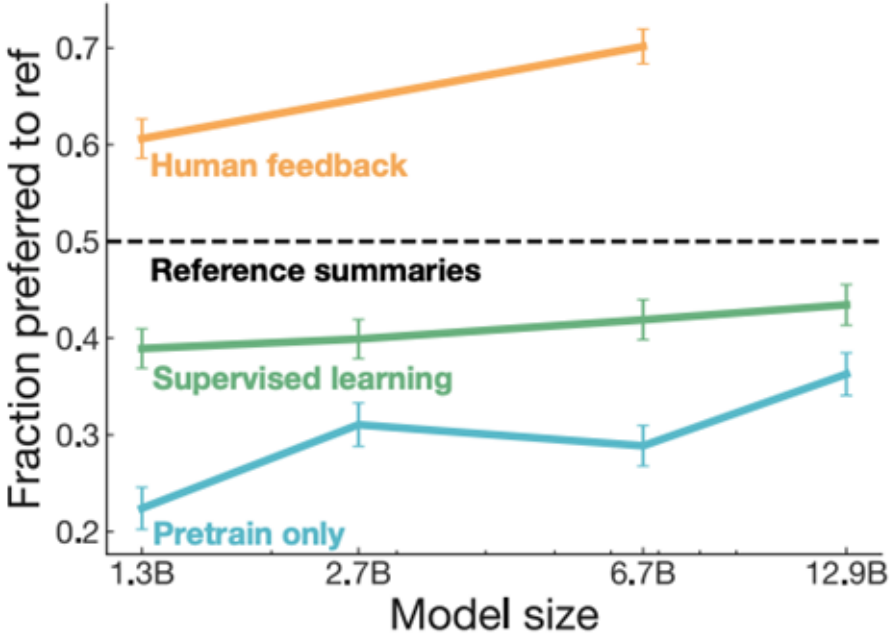
⇒ Proximal Policy Optimization (PPO)



InstructGPT, Ouyang et al, 2022

RLHF improves over supervised finetuning

Results from an earlier paper which did RLHF just for summarization



Stiennon et al, 2020

RLHF takeaways

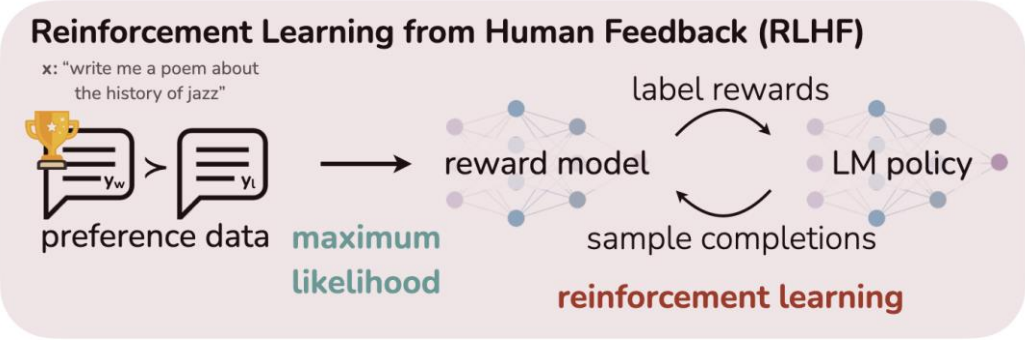
- In addition to instruction following, RLHF is used to make LMs “helpful” and “harmless” (see Bai et al, 2020 from Anthropic)
  - Which humans provide feedback is important

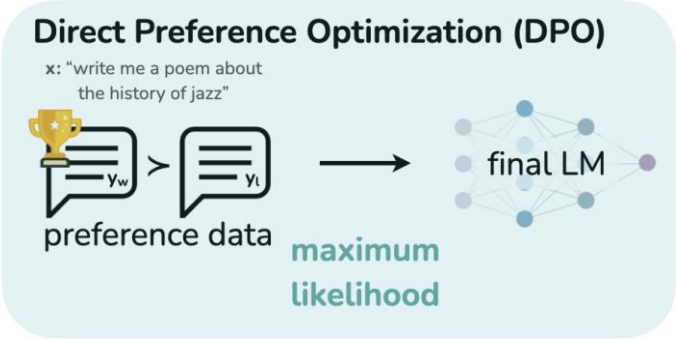
- RLHF generally leads to longer, detailed and well-formatted answers

- Many commercial LLMs today are trained using pretraining → supervised finetuning → RLHF

Alternative: DPO instead of PPO

- DPO = direct preference optimization





"increase the relative log probability of preferred to dispreferred responses"

Raifalov, Sharma, and Mitchell et al, NeurIPS 2023